Fiscal Year:	FY 2025	**Task Last Updated:**	FY 01/10/2025
PI Name:	Alwood, Joshua Ph.D.		
Project Title:	Integrated CNS Assessment in Rodent Models of Altered Gravity and Irradiation		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	650-604-1490
Organization Name:	NASA Ames Research Center		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBAS E.11: Animal Studies
Start Date:	03/01/2023	**End Date:**	02/28/2026
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Desai, Rajeev Ph.D. (McLean Hospital Corp) Kangas, Brian Ph.D. (McLean Hospital Corp) Ronca, April Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	To better understand gravity's role in life's processes, gravitational acceleration – from microgravity to hypergravity – is worthy of study in space-based and ground-based experiments across organisms. Additionally, little is known about potential interactions across the gravity continuum with exposure to ionizing radiation. In this proposed project, we aim to study the effects of gravity as a continuum, from simulated weightlessness with hindlimb unloading or from hypergravity, in combination with chronic low-dose irradiation, to assess the time course of changes in cognitive processes and associated neurotransmitter levels and circulatory biomarkers. Rats will be treated with ground analogs of spaceflight: either hindlimb unloading combined with irradiation in Aim 1, or hypergravity at 2g combined with irradiation in Aim 2. Hindlimb unloading will be achieved using custom plexiglass cages at NASA Ames Research Center (ARC). Hypergravity treatment through centrifugation will take place at NASA ARC. Low-dose irradiation will use 137Cs gamma-ray exposure. Cognitive assessment will use touchscreen and non-touchscreen. In vivo microdialysis will be performed to collect fluid for neurotransmitter assessment in key brain regions. Blood and hippocampus will be collected and assessed for complementary biomarkers. This project has the potential to advance both the state of knowledge and the state of technology to be useful in future space missions. Specifically, this project will further mechanistic underpinnings that can be associated with inflammation and oxidative stress in both the brain and circulation to determine mechanistic and translatable biomarkers integrated with gold-standard cognitive/behavioral assessment leading to peer-reviewed publications and publicly archivable data.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Assessment of various neurocognitive domains that can be meaningfully linked to changes in brain neurotransmitters and circulatory biomarkers in the brain/peripheral system as critical areas of research to uncover the risk to cognitive-sensorimotor function. The proper functioning of these neurological processes is integral for maintaining astronaut health, cognitive performance, and overall mission outcomes.
Task Progress:	In the past year, we have successfully conducted one experiment with male rats using centrifugation at 2g for 28 days and gamma irradiation, which addresses Aim 2 of the project to address intermediate and late cognitive/behavioral and neurotransmitter assessments at NASA Ames Research Center (ARC) and McLean Hospital. Additionally, assessments of cognition and neurotransmitter status have been completed on female rats at McLean Hospital.
Bibliography Type:	Description: (Last Updated: 05/25/2026)
Abstracts for Journals and Proceedings	Kangas BD, Ronca AE, Desai RI, Jenkins AR, LaMalfa KS, Puukila S, Mhatre SD, Tabares Ruiz S, Tsekov S, John ZA, Alwood JS. "Integrated neurocognitive CNS assessment in rodent models of combined altered gravity and irradiation." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024